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Olympiad angle chasing

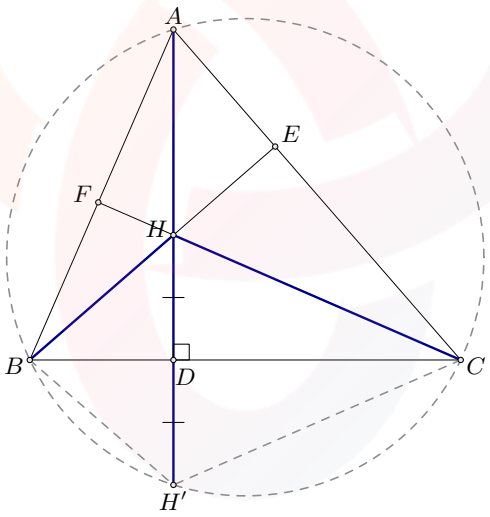
Level 4 Olympiad · Geometry

Cyclic quadrilaterals, directed angles and the standard configurations, used to write complete synthetic proofs.

By the end you can: You will be able to prove and use the converses of the cyclic and tangent criteria; You will be able to run an angle chase with directed angles that works in every configuration; You will be able to recognise orthocentre, antiparallel and incentre-excentre configurations and write BMO-standard proofs.

Lesson

Here is a fact worth owning. *In an acute triangle ABC with orthocentre H, the reflection of H in BC lies on the circumcircle.* Let BE and CF be altitudes. The right angles at E and F put A, F, H, E on a circle (the circle with diameter AH), so $\angle BHC = \angle FHE = 180^\circ - \angle A$. The reflection H' satisfies $\angle BH'C = \angle BHC = 180^\circ - \angle A$ and lies on the opposite side of BC from A, so the opposite angles of $ABH'C$ add to 180° and it is cyclic.



Nothing there was clever. We found right angles, turned them into a circle, read an angle off that circle, and turned the angle into another circle. That loop (angles give circles, circles give angles) is the engine of olympiad angle chasing. This unit is about running it reliably: knowing the converses, handling configurations without drawing six cases, recognising the standard pictures, and writing proofs a marker cannot fault.

The toolkit, with converses

Cyclic criteria. Four points A, B, C, D, no three collinear, are concyclic if and only if either of the following holds: $\angle ABD = \angle ACD$ with B, C on the same side of AD; or $\angle ABC + \angle ADC = 180^\circ$ with B, D on opposite sides of AC.
Tangent criterion. A line t through A touches circle ABC if and only if the angle between t and chord AB equals $\angle ACB$, the angle in the alternate segment.
Right angles. $\angle APB = 90^\circ$ if and only if P lies on the circle with diameter AB.

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The forward directions are the inscribed angle theorem. The converses are what make the method work, and they rest on one observation. Fix the segment AD and a circle Γ through A and D. If P is inside Γ (on a given side of AD), extend AP to meet the arc at P'; then $\angle APD$ is an exterior angle of triangle $PP'D$, so $\angle APD > \angle AP'D$. A point outside sees AD at a smaller angle in the same way. So on one side of AD, the points seeing AD at the angle given by the arc are exactly the points of that arc.

Directed angles

The weakness of these criteria is the side conditions. In a real problem a point may sit on a segment or beyond it, and the chase that says $\angle ABC + \angle ADC = 180^\circ$ in one picture says $\angle ABC = \angle ADC$ in another. The cure is to measure angles between lines, modulo 180°.

> **Directed angles.** For lines ℓ, m, let $\measuredangle(\ell, m)$ be the angle through which ℓ must be turned anticlockwise to become parallel to m, taken modulo 180°, and write $\measuredangle ABC = \measuredangle(BA, BC)$. Then $\measuredangle(\ell, m) = -\measuredangle(m, \ell)$; $\measuredangle(\ell, m) + \measuredangle(m, n) = \measuredangle(\ell, n)$; A, B, C are collinear iff $\measuredangle ABC = 0$; and $\measuredangle ABC + \measuredangle BCA + \measuredangle CAB = 0$ in every triangle.
> **Cyclic criterion:** A, B, C, D are concyclic iff $\measuredangle ABC = \measuredangle ADC$. **Tangent criterion:** t touches circle ABC at A iff $\measuredangle(t, AB) = \measuredangle ACB$.

Why does the cyclic criterion cover both cases? If B and D are on the same side of AC, triangles ABC and ADC have the same orientation, the directed angles are the ordinary ones with the same sign, and we recover equal angles. If they are on opposite sides, the orientations differ, say $\measuredangle ABC = \angle ABC$ and $\measuredangle ADC = -\angle ADC$, and equality modulo 180° means $\angle ABC + \angle ADC = 180^\circ$. One statement, both pictures.

Two warnings. First, you cannot halve: $2x = 2y$ only gives $x = y$ or $x = y + 90^\circ$, so central angles and angle bisectors need care. Route through a tangent instead. Second, directed angles forget which side of a line a point is on, so they cannot prove betweenness or compute "the angle is 40°". Use them for concyclicity, collinearity, parallels and tangency.

Configurations to know cold

The orthocentre. With feet D, E, F of the altitudes, the quadrilaterals $AFHE$, $BDHF$, $CEHD$ (diameters AH, BH, CH) and $BCEF$, $CAFD$, $ABDE$ (diameters BC, CA, AB) are all cyclic. Also $\angle BAH = 90^\circ - \angle B = \angle OAC$, so AH and AO are reflections of each other in the bisector of angle A.

Antiparallels. If X on line AB and Y on line AC make B, C, Y, X concyclic, then $\measuredangle(XY, AB) = \measuredangle YXB = \measuredangle YCB = \measuredangle ACB$, which is the angle the tangent at A makes with AB. So XY is parallel to that tangent, hence perpendicular to AO.

> **Incentre-excentre lemma.** Let line AI meet the circumcircle again at M, where I is the incentre and I_A the A-excentre. Then $MB = MC = MI = MI_A$.

Proof: M is the midpoint of arc BC, so $MB = MC$ and $\angle MBC = \angle MAC = \frac{1}{2}\angle A$. Hence $\angle MBI = \frac{1}{2}\angle A + \frac{1}{2}\angle B$, and $\angle MIB$, an exterior angle of triangle ABI, is also $\frac{1}{2}\angle A + \frac{1}{2}\angle B$. So $MB = MI$. The internal and external bisectors at B are perpendicular, so triangle IBI_A has a right angle at B; the only point of the hypotenuse II_A equidistant from B and I is its midpoint, so M is that midpoint and $MI_A = MI$.

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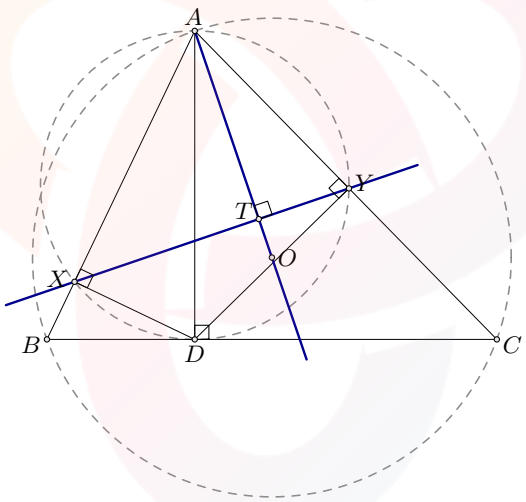
Worked examples

Example 1. Let ABC be an acute triangle with circumcentre O, and let D be the foot of the altitude from A. Let X and Y be the feet of the perpendiculars from D to AB and AC. Prove that $XY \perp AO$.

Solution.
The first instinct is to compute lengths such as $AX = AD^2/AB$. It works, but it is messy and teaches nothing. Look for right angles instead: $\angle AXD = \angle AYD = 90°$, so X and Y lie on the circle with diameter AD, and that circle hands us an angle:

$$\angle AYX = \angle ADX = 90° - \angle DAB = \angle ABC.$$

So B, C, Y, X are concyclic (exterior angle equals interior opposite angle), and by the antiparallel fact $XY \perp AO$. To finish by hand instead: triangle AOC is isosceles with apex angle $2\angle B$, so $\angle OAC = 90° - \angle B$. If AO meets XY at T, triangle ATY has angles $90° - \angle B$ at A and $\angle B$ at Y, leaving $90°$ at T.
The habit to keep: two right angles on a common segment are a circle, and that circle moves an angle from one place to another.

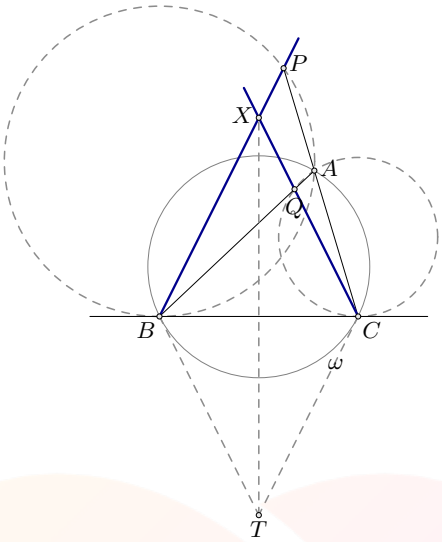


Example 2. BC is a fixed chord of a circle ω, not a diameter, and A moves on ω. The circle through A tangent to line BC at B meets line AC again at P, and the circle through A tangent to BC at C meets line AB again at Q. Prove that lines BP and CQ pass through a fixed point.

Solution.
Draw a few positions of A, on both arcs, and you will see P jump from segment AC to beyond A or beyond C. An undirected chase would need cases. With directed angles there is one argument.
By the tangent criterion in circle ABP, $\measuredangle(BC, BP) = \measuredangle BAP = \measuredangle BAC$, the last step because P lies on line AC. Now the key observation for any moving point: what stays constant? By the cyclic criterion, $\measuredangle BAC$ is the same for every point A of ω, whichever arc it is on. Call it θ. So BP is the line through B at directed angle θ to BC: a fixed line. In the same way $\measuredangle(CB, CQ) = \measuredangle CAB = -\theta$, so CQ is also a fixed line.
Two fixed lines meet at a fixed point unless they are parallel, which needs $2\theta = 0$, that is $\angle BAC = 90°$ and BC a diameter, which is excluded. We can even name the point. The tangent to ω at B makes angle $\measuredangle(BC, t_B) = -\theta$, so line BP is its mirror image in BC; likewise

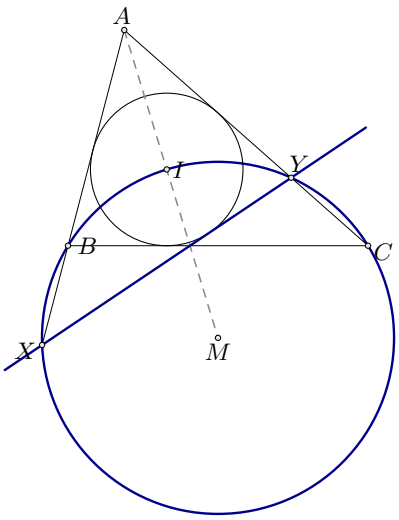
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for C. Hence the fixed point is the reflection in BC of the point T where the tangents at B and C meet.



Example 3. Let ABC be a triangle with $AB \neq AC$ and incentre I. The circle through B, I, C meets line AB again at X and line AC again at Y. Prove that XY is tangent to the incircle.

Solution.
Chasing gives $\angle AXY = \angle ACB$ quickly, since B, C, Y, X are concyclic, but tangency is a statement about distance from I, and angles at X and Y will not deliver it. So ask what the configuration knows. By the incentre-excentre lemma, circle BIC has centre M on line AI. A circle centred on a line is symmetric in that line, and reflection in AI swaps lines AB and AC. So the reflection C^* of C in AI lies on the circle and on line AB, hence is B or X. It is not B, because $AC^* = AC \neq AB$. Thus $X = C^*$ and similarly $Y = B^*$, so line XY is the mirror image of line BC. The incircle is its own mirror image and touches BC, so it touches XY. As a bonus, $AX = AC$ and $AY = AB$.
When a circle's centre lies on an axis of symmetry of the figure, reflect.



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Example 4. Let ABC be an acute triangle with circumcentre O. The circle through B, O, C meets lines AB and AC again at D and E, different from B and C. Prove that the reflection of O in DE lies on the circle through A, D, E.

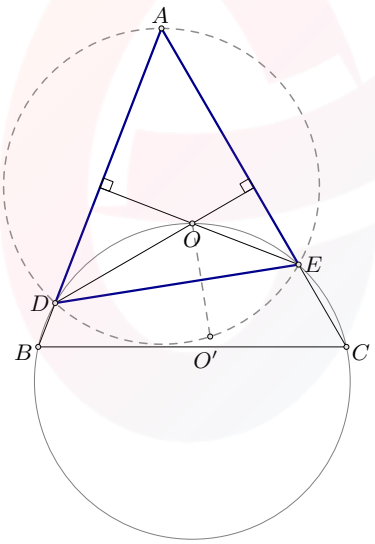
Solution.
The conclusion looks like the opening fact, so guess that O is the orthocentre of ADE, and aim for $DO \perp AE$. Depending on the shape of ABC, D may lie on segment AB or beyond B, so we use directed angles.
The trap is $\measuredangle OCB$: the familiar $90^\circ - \angle A$ comes from halving $\angle BOC = 2\angle A$, which is not allowed. Use the tangent t at C instead. Since $t \perp CO$,

$$\measuredangle(CO, CB) = \measuredangle(CO, t) + \measuredangle(t, CB) = 90^\circ + \measuredangle CAB.$$

Now circle $BOCD$ and the line ABD give

$$\measuredangle(DO, AC) = \measuredangle ODB + \measuredangle(AB, AC) = \measuredangle OCB + \measuredangle BAC = 90^\circ + \measuredangle CAB + \measuredangle BAC = 90^\circ.$$

So $DO \perp AE$, and swapping the roles of B and C gives $EO \perp AD$: O is the orthocentre of ADE. (The triangle is genuine: A cannot lie on circle BOC, as it sees BC at $\angle A$, not $2\angle A$.)
To finish without asking whether ADE is acute, redo the opening fact with directed angles. Turning both lines through 90° gives $\measuredangle DOE = \measuredangle(AE, AD) = \measuredangle EAD$. Reflection reverses directed angles, so the reflection O' satisfies $\measuredangle DO'E = \measuredangle DAE$, and A, D, E, O' are concyclic.



Writing it up

A BMO marker wants a proof that is correct in every configuration and checkable line by line. Say at the start if you use directed angles modulo 180°. Give a reason for every equality (which circle, which collinearity, which tangent). Prove a concyclicity before you use it. Deal with degenerate cases in a sentence. A diagram is a guide, never a justification.

Common traps.

- Proving the result only for your drawing when a point could lie on a segment or its extension.
- Halving directed angles, for instance with central angles or bisectors.

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- Using "it looks cyclic" in place of a criterion.
- Circular reasoning: when the claim is that P lies on a circle, do not draw P on it and read off angles. Prove an angle equality, or define P' on the circle and show $P' = P$.
- Trying to get betweenness or an actual angle size out of directed angles.

How to spot this in a contest

Reach for these tools when the conclusion is concyclicity, collinearity, parallel or perpendicular lines, or tangency, and the data are angles, right angles, tangents or circles. Mark every right angle and every tangent first: each is a circle or an angle equality waiting to be used. If you see an orthocentre, an incentre, an arc midpoint or a circumcentre, recall the standard facts before chasing blindly. If the chase stalls, draw an accurate figure and look for four points that seem concyclic but have not been joined yet; prove that as a lemma. And if a point moves, ask which directed angle stays fixed.

Practice

Try each problem before turning to the hints and solutions at the back.

1. $\star$ In triangle ABC, $\angle BAC = 58^\circ$ and $\angle ABC = 80^\circ$. Let O be its circumcentre and H its orthocentre. Find $\angle OAH$.

2. $\star\star$ In triangle ABC, $\angle A = 64^\circ$ and $\angle B = 76^\circ$. Let BE and CF be altitudes and let M be the midpoint of BC. Find $\angle EMF$.

3. $\star\star$ Let $ABCD$ be a cyclic quadrilateral with $AB = AD$, and let its diagonals meet at E. Prove that AB is tangent to the circle through B, C and E.

4. $\star\star\star$ In triangle ABC, the bisector of angle A meets the circumcircle again at M, and I is the incentre. Given that $BC = 24$, $MI = 13$ and I is the midpoint of AM, find the circumradius and the perimeter of ABC.

5. $\star\star\star$ Let ABC be an acute triangle with orthocentre H. The circle through B, H, C meets lines AB and AC again at points D and E, different from B and C. Prove that H is the circumcentre of triangle ADE.

6. $\star\star\star\star$ Let ABC be an acute triangle with circumcircle ω and circumcentre O. The tangents to ω at B and C meet at T. The line through T parallel to AB meets line AC at P. Prove that $PA = PB$.

Contest ladder

Real past problems that use this idea, easiest first. Open them on the official paper or the AoPS wiki, or online at kevinvariant08.github.io.

1. BMO1 2013/14 Problem 2 $\star\star$
2. BMO1 2015/16 Problem 2 $\star\star$
3. BMO1 2010/11 Problem 3 $\star\star$
4. AIME I 2003 Problem 10 $\star\star$

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5. BMO1 2019/20 Problem 3 ⋆⋆
6. BMO1 2023/24 Problem 3 ⋆⋆⋆
7. AIME I 2016 Problem 6 ⋆⋆⋆
8. BMO1 2014/15 Problem 5 ⋆⋆⋆
9. BMO1 2015/16 Problem 5 ⋆⋆⋆
10. USAJMO 2011 Problem 5 ⋆⋆⋆
11. USAJMO 2016 Problem 1 ⋆⋆⋆⋆
12. USAJMO 2018 Problem 3 ⋆⋆⋆⋆

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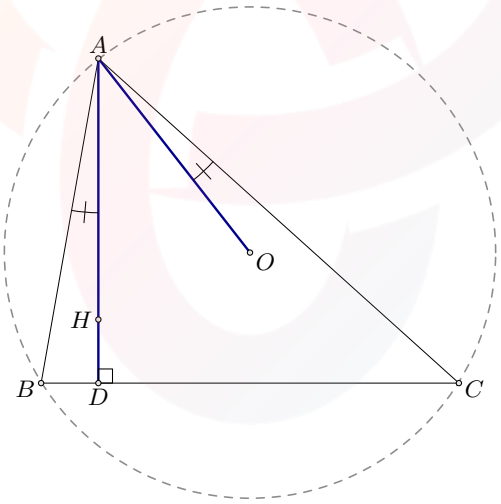
Hints

1. Work out $\angle BAH$ and $\angle OAC$ separately in terms of the angles of the triangle.
2. Where is the centre of the circle through B, C, E, F?
3. Use the converse of the alternate segment theorem: compare the angle between AB and BE with $\angle BCE$.
4. The incentre-excentre lemma tells you MB. Then look at the right-angled triangle formed by M, the midpoint of BC and B.
5. Prove $HA = HD$ by showing the base angles of triangle HAD are equal. D may be on either side of B, so directed angles are worth it.
6. First show that P lies on the circle through T, B, C. Which other notable point lies on that circle, and what does that say about the angle at P?

Solutions

Problem 1

Answer: 38°



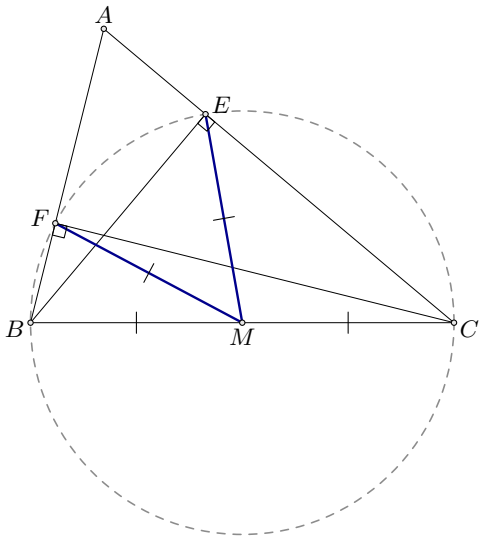
The triangle has $\angle C = 42°$, so it is acute and both O and H lie inside it. Since $AH \perp BC$, the right-angled triangle ABD (with D the foot of the altitude) gives $\angle BAH = 90° - \angle B = 10°$. Triangle AOC is isosceles with $\angle AOC = 2\angle B = 160°$, so $\angle OAC = 10°$ as well.

The rays AH and AO therefore make 10° with AB and with AC respectively, and since $10° + 10° < 58°$ they lie in that order inside angle BAC. Hence $\angle OAH = 58° - 10° - 10° = 38°$. In general $\angle OAH = |\angle B - \angle C|$.

Problem 2

Answer: 52°

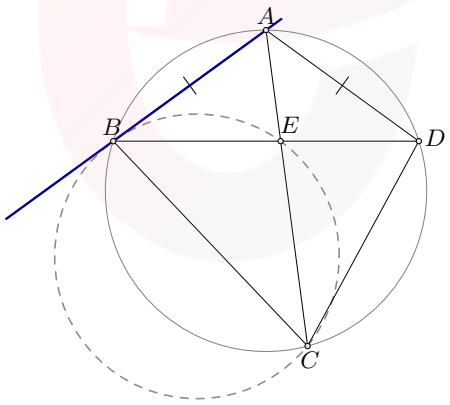
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Here $\angle C = 40^\circ$, so the triangle is acute and E, F lie on the sides. The right angles $\angle BEC = \angle BFC = 90^\circ$ put E and F on the circle with diameter BC, whose centre is M. So $MB = MC = ME = MF$.

Triangle MEC is isosceles, so $\angle MEC = \angle MCE = 40^\circ$ and $\angle EMC = 100^\circ$. Triangle MFB is isosceles, so $\angle MFB = \angle MBF = 76^\circ$ and $\angle BMF = 28^\circ$. These two angles sit on either side of $\angle EMF$ along the line BC, so $\angle EMF = 180^\circ - 100^\circ - 28^\circ = 52^\circ$. Equivalently, the angle at the centre is twice $\angle ECF = 90^\circ - \angle A = 26^\circ$.

Problem 3



Since $ABCD$ is a convex cyclic quadrilateral, the diagonal BD separates A from C, and E lies on both diagonals. Because $AB = AD$, triangle ABD is isosceles and $\angle ABD = \angle ADB$. Angles subtending the same arc AB are equal, so $\angle ADB = \angle ACB$.

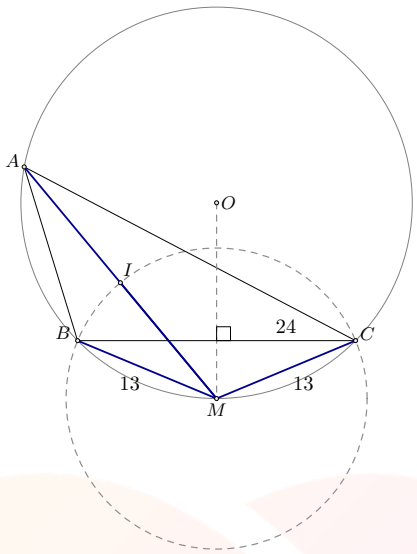
Hence $\angle ABE = \angle ABD = \angle ADB = \angle ACB = \angle ECB$, using that E lies on segments BD and AC. So the angle between line AB and the chord BE of circle BCE equals the angle $\angle BCE$ subtended by BE at C, and A and C lie on opposite sides of BE. By the converse of the alternate segment theorem, AB is tangent to circle BCE at B. (With directed angles: $\measuredangle(BA, BE) = \measuredangle ABD = \measuredangle BDA = \measuredangle BCA = \measuredangle BCE$, which is the tangent criterion.)

A useful consequence: the power of A gives $AB^2 = AE \cdot AC$.

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Problem 4

Answer: $R = \frac{169}{10}$, perimeter 72

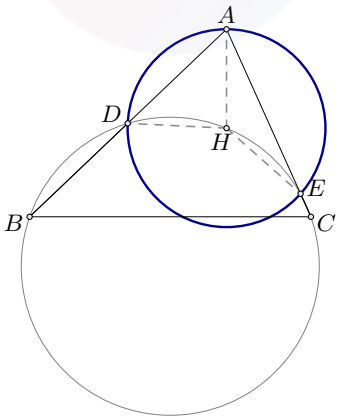


By the incentre-excentre lemma, $MB = MC = MI = 13$. As M is the midpoint of arc BC, the line OM is the perpendicular bisector of BC; let it meet BC at N. Then $BN = 12$ and, from the right angle at N, $MN = \sqrt{13^2 - 12^2} = 5$.

The centre O lies on line MN with $OM = R$, so $ON = |R - 5|$ and Pythagoras in triangle OBN gives $12^2 + (R-5)^2 = R^2$, that is $169 = 10R$, so $R = \frac{169}{10}$.

For the perimeter, $AM = 2 \cdot 13 = 26$ (which is less than the diameter 33.8, so such a triangle exists). Ptolemy's theorem in the cyclic quadrilateral $ABMC$ gives $AB \cdot MC + AC \cdot MB = AM \cdot BC$, so $13(AB + AC) = 26 \cdot 24$ and $AB + AC = 48$. The perimeter is $48 + 24 = 72$.

Problem 5



We use directed angles modulo 180°, since D may lie on segment AB or beyond B (and similarly for E). We show $HA = HD$; the same argument with B and C swapped gives $HA = HE$.

First, triangle HDA is genuine. $D \neq A$, because A is not on circle BHC: A and H are on the same side of BC and $\angle BHC = 180° - \angle A \neq \angle A$ as the triangle is acute. Also H is not on line AB, since the triangle is acute.

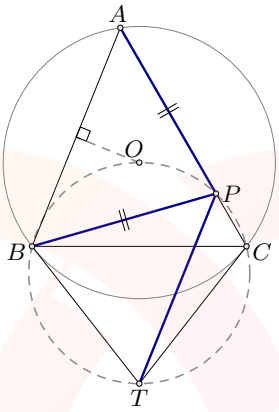
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Using that A, B, D are collinear and then that B, H, C, D are concyclic, $\measuredangle HDA = \measuredangle HDB = \measuredangle HCB$. As $CH \perp AB$, $\measuredangle HCB = \measuredangle(CH, AB) + \measuredangle(AB, BC) = 90^\circ + \measuredangle(AB, BC)$. On the other hand, as $AH \perp BC$, $\measuredangle DAH = \measuredangle(AB, AH) = \measuredangle(AB, BC) + \measuredangle(BC, AH) = \measuredangle(AB, BC) + 90^\circ$.

So $\measuredangle HDA = \measuredangle DAH$. In the triangle HDA these two directed angles are the angles at D and A, both measured in the sense of the triangle's orientation, so they equal the ordinary angles with the same sign. Two angles of a triangle that agree modulo 180° are equal, so $\angle HDA = \angle DAH$ and $HA = HD$. Together with $HA = HE$, H is equidistant from A, D, E, so it is the circumcentre of ADE.

Compare Example 4, where the circle through B, O, C produced a triangle with orthocentre O.

Problem 6



We use directed angles modulo 180°. If $CA = CB$ then the tangent at C is parallel to AB, so $P = C$ and there is nothing to prove; assume $CA \neq CB$, so $P \neq C$. Also $P \neq T$, since T is not on line AC (otherwise AC would be the tangent at C).

Step 1: T, P, B, C are concyclic. As $PT \parallel AB$ and P lies on line AC, $\measuredangle TPC = \measuredangle(AB, AC) = \measuredangle BAC$. By the tangent criterion for the tangent TB, $\measuredangle TBC = \measuredangle(BT, BC) = \measuredangle BAC$. Hence $\measuredangle TPC = \measuredangle TBC$, and T, P, B, C are concyclic.

Step 2: O lies on this circle too. The radii OB, OC are perpendicular to the tangents, so $\angle OBT = \angle OCT = 90^\circ$ and B, C lie on the circle with diameter OT. That circle is the circle TBC, so P lies on the circle with diameter OT. If $P = O$ then $PA = PB$ trivially. Otherwise $\angle OPT = 90^\circ$.

Step 3. So $OP \perp PT$, and $PT \parallel AB$, hence $OP \perp AB$. The line through O perpendicular to AB is the perpendicular bisector of AB, so P lies on it and $PA = PB$. ■

The same argument shows that P lies on circle BOC, which links this problem to Example 4.

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